

Emanuele Bugliarello

✉ emanuele@di.ku.dk ·  emanuelebugliarello ·  e-bug.github.io ·  e-bug

Research Interests

Multi-modal and multilingual natural language understanding, representation learning, transfer learning

Education

PhD in Computer Science 2019–present
University of Copenhagen Copenhagen, Denmark
Advisors: Desmond Elliott, Anders Søgaard

MSc in Communication Systems, Data Analytics specialization 2015–2018
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)

BEng in Electronics and Information Engineering 2013–2018
Tongji University (同济大学) Shanghai, China
Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30

BSc in Telecommunications Engineering 2012–2015
Politecnico di Torino Turin, Italy
GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 *Cum Laude*

Experience

Research Intern Upcoming
Spotify Research London, UK
On Multi-objective Sequential Recommendations

Research Associate with Prof. Naoaki Okazaki 10/2018–09/2019
Okazaki Lab @ Tokyo Institute of Technology (東京工業大学) Tokyo, Japan
On Neural Machine Translation

- Design of an effective syntax-aware self-attention mechanism for the Transformer model
- Proposing cross-mutual information for measuring difficulty of neural translation

Visiting Scholar with Prof. Rui Fan and Prof. Dan Alistarh 04/2018–07/2018
ShanghaiTech University (上海科技大学) Shanghai, China
Large-Batch Algorithms for Deep Learning at Scale

- Implementation of state-of-the-art algorithms for large-batch training of Convolutional Neural Networks
- Evaluation of large-batch training algorithms performance on different datasets to assess their generality

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot 09/2017–03/2018
Technicolor AI Lab Palo Alto, CA, USA
Intelligent VFX Scheduling

- Recommending employees to visual effects (VFX) departments with novel matrix factorization models

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller Summer 2017
Swiss National Supercomputing Centre (CSCS) Lugano, Switzerland
Distributed TensorFlow: A performance evaluation

- Scalability study of distributed TensorFlow using InceptionV3 for image classification on ImageNet

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308)

Spring 2017

Publications

3. **It's Easier to Translate *out of* English than *into* it: Measuring Neural Translation Difficulty by Cross-Mutual Information**

E. Bugliarello, S.J. Mielke, A. Anastasopoulos, R. Cotterell, N. Okazaki
In *ACL 2020*

2. **Enhancing Machine Translation with Dependency-Aware Self-Attention**

E. Bugliarello, N. Okazaki
In *ACL 2020*

1. **Matrix Completion in the Unit Hypercube via Structured Matrix Factorization**

E. Bugliarello, S. Jain, V. Rakesh
In *IJCAI 2019*

Projects

Evaluating the evolution of Wikipedia's navigability

Semester project with Prof. Robert West

Spring 2017
Data Science Lab @ EPFL

Investigate Wikipedia's navigability over the course of its history, based on the English Wikipedia's complete edit history

- Poster accepted at *Wiki Workshop, WWW 2018*

Solarity: Solar Clarity for Swiss Households

Applied Data Analysis (CS-401)

Fall 2016

EPFL

Based on historical weather data, predict solar energy production in the coming years to provide information on initial investment, break-even time, and potential savings for domestic solar installations

- 2nd best class project award (over 60+ projects)

Twitter Sentiment Classification

Machine Learning (CS-433)

Fall 2016

EPFL

Examine the accuracy of different learning algorithms in determining sentiments associated to tweets

Skills

Programming

- Python, C, Java, Matlab, Java for Android
- NumPy, Pandas, scikit-learn, CUDA
- PyTorch, TensorFlow
- Spark, Hadoop MapReduce, MPI
- SQL, MySQL, Mongo

Languages

- Italian (native)
- English (fluent – TOEFL iBT 109)
- French (intermediate)
- Chinese (intermediate – HSK III)
- Japanese (basic)

Fellowships, Honors, and Awards

University of Copenhagen Marie Skłodowska-Curie TALENT Doctoral Fellowship	06/2019
EPFL PhD Program in Computer Science Fellowship [offer declined]	05/2019
The Japanese Government (MEXT) Scholarship [offer declined]	01/2019
EPFL TCP/IP 2015-2016 Research Exercise Award	12/2015
EIT Digital Master School Excellence Scholarship [offer declined]	04/2015
PoliTong Merit Scholarship	07/2013
MIUR National Register of Excellences	06/2012